



University of Rwanda

College of Science and Technology

Bachelor of Computer Science

Web Application for Monitoring Trash Collection Schedules and Promoting Efficient Solid Waste Management

Module: Research Methodology

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Chapter 1: Introduction

1.1. Introduction

1.1.1 Global Context of Solid Waste Management

Rapid global urbanization and explosive population growth continue to place immense strain on traditional municipal solid waste management (SWM) systems across the globe. According to the World Bank, global waste generation is expected to increase by 70% by 2050 unless urgent interventions are made [10]. Historically, waste collection has relied on fixed, rigid pickup schedules that do not dynamically adapt to actual waste accumulation rates. This foundational misalignment frequently results in overflowing garbage bins, increased greenhouse gas emissions from rotting organic matter, and a notable communication gap between sanitation authorities and citizens. In developed nations, the integration of smart technologies has mitigated many of these issues; however, developing nations are still grappling with manual, inefficient, and costly logistical frameworks [2].

1.1.2 Regional Context: Africa and Urban Sprawl

In the context of the African continent, the challenge of municipal solid waste management is exacerbated by informal settlements, a lack of adequate infrastructure, and limited municipal budgets. Cities across Sub-Saharan Africa are experiencing urbanization rates that outpace infrastructural development. Consequently, local governments often spend up to 50% of their operational budgets purely on the logistics of waste collection, yet collection rates in many areas remain below 60% [10]. The reliance on antiquated, paper-based tracking and the lack of real-time monitoring mean that resources are deployed inefficiently.

1.1.3 Local Context: Kigali City, Rwanda

In Rwanda, and particularly in its rapidly growing capital urban center, Kigali, the population has grown to over 4 million inhabitants. This population boom has led to a surge in municipal solid waste generation, estimated currently between 400 and 800 tons per day. The City of Kigali (CoK) has made globally recognized and commendable strides in city cleanliness, often celebrated as one of the cleanest cities in Africa. Private companies, notably COPED (Company for Protection of Environment and Development) and Agruni Group, are contracted to manage waste collection across various sectors (Gasabo, Nyarugenge, and Kicukiro).

However, despite these accolades, the reliance on manual tracking and predefined collection routes exacerbates operational bottlenecks. Trucks frequently expend expensive imported fuel visiting empty

bins in certain neighborhoods while ignoring overflowing ones in higher-density sectors. Ultimately, this inefficient system puts immense pressure on the primary Nduba dumpsite, which currently struggles with capacity and environmental safety concerns [5].

1.1.4 Proposed Digital Intervention

To address these growing logistical inefficiencies, our project proposes the design and implementation of a comprehensive web-based application to enable real-time tracking, optimize logistical coordination, and empower citizen engagement [3]. By leveraging modern digital tools, including smart scheduling algorithms and interactive citizen reporting interfaces, this solution seeks to modernize the entire SWM lifecycle in Kigali. It aligns seamlessly with Rwanda's broader "Smart Cities" master plan and the Vision 2050 goals [11], ensuring higher municipal accountability, significantly lowering operational and fuel costs, and fostering better resource allocation across Kigali's diverse municipalities. The proposed system is a user-friendly web application designed to improve waste management by combining key services in one platform. It includes a trash collection tracker to prevent missed pickups, an educational hub for sustainable practices, and a recycling transaction feature that connects users with junkshops. With mobile accessibility, the system promotes convenience, responsible waste disposal, and environmental awareness.

1.2. Problem Statement

Despite significant advancements in urban planning and a strong political will directed toward environmental sustainability, effective solid waste management remains a critical operational hurdle in Kigali City. The core issue lies in the continued reliance on outdated, static collection methodologies. Sanitation vehicles operated by private contractors currently navigate fixed, predefined routes without any real-time intelligence regarding the actual fill level of neighborhood bins.

This "blind" logistical approach leads to severe operational inefficiencies. Municipal resources such as

fuel, labor, vehicle maintenance, and time are frequently wasted on servicing empty or half-full bins. Conversely, bins in high-traffic commercial zones (such as Nyabugogo and Kigali City Market) or densely populated peri-urban areas overflow long before their scheduled pickup time. This discrepancy not only drains municipal and private budgets but also contributes to unnecessary carbon emissions from heavy-duty collection trucks circling the city inefficiently [4].

Furthermore, there is a pronounced lack of transparency, accountability, and communication between municipal waste companies and the general public. Residents frequently miss unannounced collection trucks, face unexplained delays during the rainy season, and have no streamlined, official digital medium to report missed collections, illegal dumping incidents, or to request specialized pickups for bulky or hazardous waste.

In areas adjacent to the landfill, such as Kinyinya and Nduba sectors, poor waste management and delayed collections have been linked to negative livelihood impacts, including the proliferation of vectors (mosquitoes, rodents) and the spread of waste-borne diseases. The complete absence of a centralized, data-driven analytics platform prevents local authorities (like the City of Kigali and REMA) from strategically allocating resources [5], ultimately resulting in creeping environmental degradation, public health hazards, and massive, unnecessary financial expenditures [2].

1.3. Statement of Objectives of the System

1.3.1 General Objective

The primary overarching objective of this project is to design, develop, and deploy a smart, web-based application localized for Kigali City that optimizes municipal solid waste management by providing real-time schedule monitoring, facilitating active community reporting, and enabling data-driven logistical route planning for municipal waste collection companies.

1.3.2 Specific Objectives

1. **To build a user-centric citizen portal:** Develop an intuitive, mobile-responsive web interface where Kigali citizens can easily access real-time, updated trash collection schedules and report overflowing bins or uncollected waste directly to their respective service providers.
2. **To optimize fleet route logistics:** Implement a secure data-tracking and visualization

dashboard for municipal authorities and private companies (like COPED/Agruni) to visualize waste accumulation hotspots dynamically and plan fuel-efficient, algorithmically optimized collection routes.

3. **To bridge the civic communication gap:** Establish a transparent, two-way communication channel connecting waste management service providers with local residents, featuring push notifications and alerts, to significantly reduce illegal dumping and missed collections.
4. **To support strategic administrative decision-making:** Provide foundational, exportable data analytics that allow local authorities (City of Kigali, MININFRA) to monitor SWM efficiency across different sectors, track the volume of waste destined for the Nduba landfill, and continuously improve service delivery.

1.3.3 Research Questions

To achieve the above objectives, this study seeks to answer the following key questions:

1. What are the primary bottlenecks and inefficiencies in the current manual solid waste collection scheduling system used in Kigali City?
2. To what extent can a web-based reporting and scheduling application improve the communication between Kigali residents and waste collection companies?
3. How can the integration of digital routing algorithms reduce the operational costs (e.g., fuel consumption and time) for private waste management contractors in Rwanda?
4. What are the usability and acceptance rates of digital SWM tools among the general population of Kigali?

1.4. Scope and Limitations of the Study

1.4.1 Geographical Scope

The research and subsequent software deployment will be exclusively limited to the geographical boundaries of Kigali City, Rwanda. Specifically, testing and user acceptance will focus on a sampling of three distinct sectors representing different urban densities: Remera (commercial/residential mix), Kimisagara (high-density residential), and Kinyinya (peri-urban).

1.4.2 Technical Scope

The technical scope involves the development of a web-based application accessible via standard desktop and mobile browsers. It will not include the deployment of physical IoT hardware (such as smart bin sensors), due to budget constraints. Instead, the system will rely on a "crowdsourced" citizen-reporting model and algorithmic scheduling to simulate hardware inputs [7].

1.4.3 Limitations

- **Internet Accessibility:** The system's effectiveness relies on citizen access to smartphones and internet connectivity, which may exclude certain lower-income demographic segments.
- **Time Constraints:** The four-month development and testing window restricts the ability to conduct long-term longitudinal studies on cost savings.

1.5. Significance of the Study

The successful completion and implementation of this research project will yield significant benefits across multiple tiers of Rwandan society:

1. **For the Citizens of Kigali:** Residents will experience a cleaner environment, reduced health risks from overflowing waste, and a modernized, transparent way to interact with sanitation services. Predictable schedules will allow households to manage their waste more effectively.
2. **For Waste Management Companies (e.g., COPED, Agruni):** The data-driven backend will allow these companies to transition from static to dynamic routing. This will lead to direct financial savings through reduced fuel consumption, lowered vehicle wear-and-tear, and optimized labor hours.
3. **For the Government and Municipal Authorities:** The City of Kigali, the Rwanda Environment Management Authority (REMA), and the Ministry of ICT (MINICT) will gain a valuable macro-level overview of waste generation trends. This data will be crucial for urban planning, budgeting, and managing the intake capacity at the Nduba dumpsite.
4. **For the Academic and Tech Community:** This study provides an empirical framework for software-driven civic tech in East Africa. It contributes to the existing body of knowledge regarding how purely software-based interventions (without expensive hardware) can leapfrog infrastructural deficits in developing nations [4].

Chapter 2: Literature Review

2.1. Literature Review

2.1.1 Theoretical Framework: Technology Acceptance Model (TAM)

This study is grounded in the Technology Acceptance Model (TAM), formulated in [9].

TAM suggests that the adoption of any new technological system is primarily determined by two factors: *Perceived Usefulness* (the degree to which a person believes the system will enhance their daily life) and *Perceived Ease of Use* (the degree to which using the system is free of effort). In the context of Kigali, for a SWM web app to succeed, it must be exceptionally intuitive for the average citizen and demonstrably useful (i.e., it must result in actual trash pickup) to ensure sustained engagement [9].

2.1.2 Digital Transformation in Municipal SWM

Recent academic literature highlights a massive global shift away from manual sanitation workflows toward automated, data-driven frameworks. Studies on digital transformation in waste management emphasize that integrating digital governance tools (like WebGIS and citizen portals) transitions cities toward "zero-waste" models. Traditional methods inevitably result in excessive fuel consumption and labor inefficiencies. Digital interventions, conversely, utilize real-time data to lower costs and minimize environmental footprints, acting as a disruptive innovation vital for sustainable urban development [2].

2.1.3 SWM in the Context of Kigali City

Research specific to Kigali indicates a strong governmental push toward technological integration. The Ministry of ICT and Innovation (MINICT), alongside the City of Kigali, recently launched Smart Waste Management pilot projects incorporating solar-powered, IoT-enabled dustbins and CCTV monitoring in specific high-profile zones (like the Car-Free Zone) [11]. However, while commercial hardware solutions are expanding in public spaces, there remains a critical, documented gap in software ecosystems that connect ordinary residential citizens to these advancements. Localized strategic management principles dictate that adopting digital platforms tailored to the socio-economic and cultural realities of Kigali is essential for long-term operational sustainability [5].

2.1.4 Enhancing Community Participation through Civic Tech

Contemporary waste management models view citizens not merely as passive service recipients, but as active, vital stakeholders in urban hygiene. Providing residents with web-based applications to view schedules, report hazards, and track the resolution of issues promotes deep civic responsibility. Studies from similar developing urban centers (e.g., Nairobi, Accra) show

that participatory software platforms drastically reduce instances of illegal dumping. By fostering a sense of ownership and holding both the public and authorities accountable via a digital paper trail, overall municipal hygiene improves significantly [3], [6].

2.1.5 Logistical Route Optimization and Geographic Information Systems

Geographic Information Systems (GIS) form the technical backbone of modern smart waste routing. By tracking user reports and visualizing geo-tagged data on a digital map, planners can assess the spatial dynamics of municipal waste in real-time [1]. This geographic awareness allows for an "on-demand" or predictive dispatch model. Instead of visiting every street, algorithms can calculate the shortest, most fuel-efficient paths that only target full bins or reported areas, consequently reducing a city's carbon footprint and mitigating rush-hour traffic congestion [7].

2.1.6 Artificial Intelligence in Waste Classification

Though slightly beyond the immediate scope of this specific software implementation, advancements in AI and Machine Learning have introduced automated waste categorization to SWM. Systems equipped with computer vision can analyze user-submitted photos to classify waste types, suggest recycling protocols, and flag hazardous materials, accelerating municipal response times [8]. Our proposed system will lay the database groundwork for future AI integration by standardizing the way user reports are structured and stored.

2.1.7 Summary Table of Research Gaps and Project Innovation

Author(s) & Year	Title/Focus	Key Contribution	Identified Gaps in Literature	Our Proposed Solution
Akpotuzor (2020)	Web-GIS for Solid Waste Monitoring	Developed a GIS-based tracking system for bin locations.	Lacked interactive scheduling for end-users/citizens.	Incorporates user-facing schedules and bidirectional live issue tracking.
Luiz Gustavo Francischinelli	Digital Transformation in	Highlighted digital governance for	Addressed high cost of	Uses a hybrid crowdsourcing software

Rittl, Atiq Zaman, Francisco Henrique de Oliveira(2025)	SWM	zero-waste cities.	hardware/IoT implementation but offered no software alternative.	model relying on citizen reporting to bypass hardware costs.
MINICT Rwanda (2024)	Smart Waste Pilot Kigali	Deployed IoT bins in specific public markets.	Focused strictly on commercial hardware; lacks residential software integration for suburbs.	Bridges the gap by providing an accessible web portal specifically for residential and peri-urban reporting.
Nsabimana (2012)	Strategic Management in Rwanda	Identified need for localized strategic management tools.	Focused on general business operations; not applied to SWM or civic tech.	Applies strategic efficiency principles specifically to a dedicated SWM web portal for local contractors.
Dundi Harini,Gannoji Chikitha,Shaik Fatima, Gnyana Deepa (2024)	ML-Based Waste Suggestions	Applied AI to classify waste and suggest reuse.	Focused on individual/home waste; lacked municipal collection logistics.	Integrates reporting algorithms directly into municipal workflow and contractor dashboards.

Chapter 3: Methodology

3.1. Study Methods and Methodology

3.1.1 Research Philosophy and Design

The project will utilize a **Design Science Research Methodology (DSRM)** , heavily combining both qualitative and quantitative approaches. DSRM is specifically tailored for IT research, as it

focuses on the creation and evaluation of an innovative IT artifact (the web application) intended to solve an identified organizational problem. The focus will be on diagnosing current bottlenecks in Kigali's SWM processes (specifically analyzing the workflow of local companies and Nduba landfill deposits) and engineering a viable, scalable software solution.

3.1.2 Population and Sampling Strategy

The target population for this study includes three primary groups:

1. **General Citizens:** Residents of Kigali City.
2. **Waste Management Personnel:** Drivers, logistics managers, and administrators at companies like COPED.
3. **Municipal Authorities:** Officials from the City of Kigali sanitation department.

A **stratified random sampling technique** will be employed to select 150 citizens across three varied sectors (Remera, Kimisagara, Kinyinya) to ensure diverse socio-economic representation. Furthermore, **purposive sampling** will be used to select 10 key informants from the waste management companies and municipal authorities for in-depth interviews.

3.1.3 Data Collection Instruments

Data will be sourced by triangulating multiple instruments to ensure validity:

- **Structured Questionnaires:** Distributed electronically (via Google Forms) to the citizen sample to gauge their current frustrations with SWM, their internet accessibility, and their willingness to use a web portal.

3.1.4 Software Development Methodology

The application will be developed utilizing an **Agile (Scrum) framework**. This methodology aligns with modern, fast-paced software development practices. The team will work in two-week "sprints."

Crucially, the research team will employ modern collaborative development techniques such as peer programming and continuous integration/continuous deployment (CI/CD) workflows, skills commonly emphasized and provided in free, global, remote software development. Utilizing these industry-standard collaborative paradigms ensures that the

codebase remains clean, version control (via Git/GitHub) is strictly maintained, and the team can rapidly iterate based on user feedback during the testing phases.

3.2. System Analysis and Design

The System Analysis and Design (SAD) process for this project utilized Object Oriented Analysis and Design Methodology (OOADM), aligned with the overall Agile (Scrum) methodology adopted for development. This chapter formally outlines the analysis and design process for the Web Application for Monitoring Trash Collection Schedules and Promoting Efficient Solid Waste Management in Kigali City, Rwanda. It moves from an analysis of the current operational environment to the specification of the proposed technical solution, covering architectural design, detailed requirements, data structuring, and project feasibility.

3.2.1 Analysis of the Existing System (Current Waste Management Challenges)

The current solid waste management (SWM) framework in Kigali City, managed by contractors like COPED and Agruni Group, is a predominantly manual system reliant on static, predefined collection routes. This methodology presents several critical logistical and public health challenges:

- **Operational Inefficiency and Cost:** Sanitation vehicles operate without real-time intelligence, resulting in "blind" collection runs. Significant municipal resources, including fuel, labor, and maintenance, are frequently wasted servicing empty or low-fill bins. This inefficiency contributes to unnecessary carbon emissions.
- **Service Discrepancies:** While some areas receive excessive service, high-density commercial (e.g., Nyabugogo) or peri-urban sectors often experience overflowing bins well before the scheduled pickup. This discrepancy fosters unsanitary conditions, contributes to the proliferation of vectors (mosquitoes, rodents), and increases the risk of waste-borne diseases, particularly near areas like Kinyinya and Nduba.
- **Lack of Transparency and Communication:** There is an absence of a streamlined, official digital channel for bidirectional communication. Citizens cannot easily report

missed collections or illegal dumping, and contractors cannot effectively communicate unannounced delays, leading to public dissatisfaction and reduced accountability.

The existing system's reliance on paper-based tracking fails to provide the data analytics necessary for the City of Kigali (CoK) or REMA to perform strategic resource allocation and manage the capacity of the Nduba dumpsite.

3.2.2 Proposed System Overview

The proposed web application is a comprehensive digital intervention designed to shift Kigali's SWM from a static, reactive model to a dynamic, data-driven, and proactive one. The system directly addresses the core problem of logistical blindness and communication gap by:

- **Enabling Dynamic Routing:** The Citizen Module facilitates crowdsourced, geo-tagged reporting of overflowing waste. This real-time data feeds into the Contractor Dashboard, allowing for the algorithmic calculation of optimized, fuel-efficient routes that prioritize high-need areas instead of fixed routes. This fulfills the specific objective of optimizing fleet route logistics.
- **Fostering Transparency and Engagement:** The citizen portal provides reliable, real-time collection schedules and notifications, empowering residents and bridging the civic communication gap.
- **Supporting Strategic Oversight:** The Administration Portal centralizes SWM data, providing authorities with exportable analytics to monitor contractor performance (e.g., response times) and trends in waste generation for long-term planning.

The system operates as a **three-tier platform** targeting Citizens, Contractors, and Municipal Administrators.

3.2.3 System Architecture

The application adopts a robust **Three-Tier Architecture** conforming to the Model-View-Controller (MVC) design pattern for enhanced scalability and maintainability.

- **Presentation Tier (Frontend):** Developed using **React.js** and styled with **Tailwind**

CSS. This ensures a Single Page Application (SPA) experience, promoting responsiveness and a mobile-first design, which is essential given the primary method of access for Kigali residents.

- **Application Tier (Backend/Business Logic):** Implemented using **Node.js** with the **Express framework**. This stack manages RESTful API requests, performs user authentication (via **JWT**), and executes the core business logic, including the route optimization algorithm and schedule management.
- **Data Tier (Database/Storage):** **PostgreSQL** serves as the primary relational database. Crucially, it incorporates the **PostGIS extension** to efficiently store, index, and query the geographical coordinates (geo-tags) submitted with citizen reports, which is vital for the GIS mapping and route calculations. High-volume, non-structured data like user-uploaded photos are stored separately in an **AWS S3 bucket** to maintain database performance.

The prototype is intended to be hosted on scalable cloud infrastructure such as Heroku or Vercel.

3.2.4 Use Case Analysis

The system facilitates interactions among three distinct actors, each possessing specific permissions and performing a defined set of actions relevant to the SWM process:

Actor	Description	Primary Use Cases (Interactions)
Citizen (Kigali Resident)	The end-user provides input and consumes schedule information.	Register/Authenticate; View Collection Schedule; Submit Geo-tagged Waste Report (photo-assisted); Receive Service Notifications; Track Report Resolution Status.
Contractor/Logistics Manager	Personnel from private SWM companies (COPED, Agruni) responsible for operational dispatch and fleet management.	Login/Authenticate; View Real-time Waste Report Heatmap; Generate Dynamic Optimized Route; Assign Route to Driver; Mark Collection as Completed; Generate Fleet Performance Reports.

Municipal Administrator	Officials from the City of Kigali or REMA responsible for oversight, regulation, and strategic planning.	Login/Authenticate; View SWM Efficiency Metrics; Manage Master Data (Sectors, Tariffs); Export Historical Data and Analytical Reports.
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3.2.5 Data Flow Explanation

The data flow within the system is initiated by citizen reports and culminates in logistical actions, creating a continuous feedback loop:

1. **Input:** A **Citizen** submits a *Waste Report* via the mobile portal. The system captures five critical data points: User ID, Incident Type (e.g., overflow), Timestamp, Photo URL (from S3), and the **GPS Geo-tag** (PostGIS location).
2. **Processing:** The **Node.js/Express backend** processes the incoming API request. It validates the data, updates the *WasteReports* table, and immediately integrates the geo-tag into the **GIS layer**. The system continuously analyzes the density of open reports to create a dynamic "hotspot" map visible on the Contractor Dashboard.
3. **Output (Action):** The **Contractor/Logistics Manager** views the heatmap and triggers the **Route Optimization Algorithm**. This algorithm uses the clustered hotspot data and current vehicle fleet location to compute the shortest, most fuel-efficient paths that only target full bins or reported areas, outputting the sequence of pickup locations (the *Optimized Route*) to the driver's interface.
4. **Feedback Loop:** Upon completion of the route, the driver marks the reports as *Resolved*. This status change triggers a notification (e.g., SMS/push alert) back to the original **Citizen** who submitted the report, confirming resolution and closing the communication loop. The resolved data is logged as historical records for administrative analysis.

3.2.6 Database Design Description

The data layer is built on PostgreSQL, utilizing its robust relational capabilities and the essential PostGIS extension for spatial data handling. The conceptual Entity-Relationship Model (ERD) is based on five primary entities:

- **Users:** Stores system access credentials, roles (Citizen, Contractor, Admin), contact

information, and linkage to their geographical sector.

- **WasteReports:** The core operational table. Stores all reported incidents, including the spatial coordinate (PostGIS *geometry* type), status (*Pending*, *Resolved*), incident details, and the foreign key link to the reporting User.
- **CollectionSchedules:** Manages the fixed and dynamic schedules, detailing sector-specific pickup times. This allows the system to send proactive schedule notifications and track deviations.
- **Routes:** Stores generated and executed routes, including the sequence of pickup locations to be serviced, estimated travel distance/time, and the assigned *Vehicle* ID.
- **Vehicles/Fleet:** Manages the assets of the private contractors, storing vehicle registration, capacity, fuel efficiency metrics, and driver assignments.

3.2.7 Functional Requirements (The System Shall...)

These requirements define the necessary behaviors of the system:

- **F.R. 1.1:** The system shall enable secure user authentication and authorization based on defined roles (Citizen, Contractor, Admin).
- **F.R. 1.2 (Citizen):** The system shall display the scheduled collection dates and times for the user's registered sector/cell.
- **F.R. 1.3 (Citizen):** The system shall enable the submission of a *Waste Report* that includes mandatory GPS coordinates and an optional photo attachment.
- **F.R. 2.1 (Contractor):** The system shall present a real-time geographical heatmap visualization of open *WasteReports*.
- **F.R. 2.2 (Contractor):** The system shall calculate and suggest the optimal, shortest-path route for collection based on the real-time reported hotspots.
- **F.R. 2.3 (Contractor):** The system shall allow authorized personnel to update the status of individual reports from *Pending* to *Resolved*.
- **F.R. 3.1 (Communication):** The system shall send push notifications or SMS alerts to citizens regarding unannounced service delays or schedule changes.
- **F.R. 3.2 (Admin):** The system shall generate exportable efficiency and performance

reports for the Municipal Administrator, detailing response times and service volume per sector.

3.2.8 Non-Functional Requirements

These requirements specify the quality attributes of the system:

- **N.F.R. 1 (Security):** All user passwords shall be stored using industry-standard **bcrypt hashing**. API access shall be strictly controlled using **JSON Web Tokens (JWT)** for session management. The system shall be tested to prevent common vulnerabilities like SQL Injection and XSS.
- **N.F.R. 2 (Performance):** The primary user interface shall load completely within **three seconds** under typical Rwandan mobile network conditions (3G/4G, e.g., MTN, Airtel). The route optimization algorithm shall execute and deliver a suggested route within five seconds of the request.
- **N.F.R. 3 (Usability/Localization):** The interface shall adhere to a **mobile-first design** and be highly intuitive. The system must support bilingual content for **English and Kinyarwanda** to ensure maximum adoption.
- **N.F.R. 4 (Scalability):** The chosen architecture (Node.js, PostgreSQL/PostGIS) and cloud hosting (Heroku/Vercel) must be capable of handling an anticipated load of **at least 1,000 concurrent citizen reports** during peak usage without degradation of service.

3.2.9 Feasibility Study

A formal feasibility analysis was conducted based on three dimensions to confirm the viability of the project:

- **Technical Feasibility:** The chosen technology stack (React, Node.js, PostgreSQL/PostGIS) utilizes stable, open-source, and widely supported frameworks, minimizing technical risk. Crucially, the solution avoids the complexity and budget strain of deploying physical IoT hardware, relying instead on the established and technically accessible **crowdsourced reporting model**. The research team possesses the necessary skills in these technologies.

- **Economic Feasibility:** As an academic project, development labor costs are contained. The total estimated budget is modest (45,000 RWF). The economic justification rests on the significant potential for **cost savings** for private contractors through optimized routing and reduced fuel consumption, confirming a high Return on Investment (ROI).
- **Operational Feasibility:** The system aligns directly with Kigali's SWM goals and integrates smoothly with existing contractor workflows, replacing only the outdated paper-based tracking. Its usability focus (Kinyarwanda support, mobile-first design) is expected to ensure high user acceptance, aligning with the **Technology Acceptance Model (TAM)** principles of *Perceived Ease of Use*.

3.2.10 Constraints and Assumptions

The project's successful delivery is subject to clearly defined constraints and operational assumptions:

- **Project Constraints:**
 - **Geographical Scope:** User acceptance testing and system deployment are strictly limited to the pilot sectors: Remera, Kimisagara, and Kinyinya.
 - **Time:** The rigid **four-month development and testing window** prohibits the execution of a long-term longitudinal study on environmental or financial impacts.
 - **Resource:** The system **excludes the use of physical IoT bin sensors**, relying entirely on citizen-reported data for real-time intelligence.
- **Operational Assumptions:**
 - **Data Integrity:** It is assumed that the majority of citizens will submit truthful and accurate geo-tagged reports, ensuring the integrity of the data used for route optimization.
 - **Infrastructure:** The feasibility of the system relies on the assumption that a sufficient percentage of the target population possesses reliable **smartphone access and internet connectivity (3G/4G)**.
 - **API Access:** It is assumed that necessary mapping services (e.g., Google Maps

API or Mapbox) are accessible and affordable for the implementation of the core GIS functionality.

3.3. Research Plan and Detailed Timeline

The project development lifecycle spans precisely four months, structured chronologically from the beginning of February 2026 to the final defense on May 25, 2026. This timeline utilizes a rigorous Agile approach to ensure all deliverables are met.

Phase & Specific Task / Activity	February 2026	March 2026	April 2026	May 2026
PHASE 1: Research, Planning & Requirements Engineering				
Extensive Literature Review & Theoretical Framework Setup	Feb 1 - Feb 7			
Kigali Context Study & Ride-along Observations	Feb 8 - Feb 14			
Questionnaire Distribution & Citizen User Surveys	Feb 15 - Feb 21			
Requirements Analysis & Project Proposal Defense	Feb 22 - Feb 28			
PHASE 2: System Architecture & UI/UX Design				
Database Schema Design (PostgreSQL/ERD Diagrams)		Mar 1 - Mar 7		
UML Modeling (Use Case, Class, Sequence Diagrams)		Mar 8 - Mar 14		
High-Fidelity UI/UX Wireframing (Figma)		Mar 15 - Mar 22		
Prototype Approval & Setup of CI/CD Git Workflows		Mar 23 - Mar 31		

PHASE 3: Software Implementation (Agile Sprints)				
Sprint 1: Frontend Development (Citizen Dashboard & Auth)			Apr 1 - Apr 10	
Sprint 2: Backend Development (Node.js REST API & DB)			Apr 11 - Apr 20	
Sprint 3: Logic Integration (Geo-tagging & Admin Dashboard)			Apr 21 - Apr 30	
PHASE 4: Quality Assurance, Deployment & Delivery				
Alpha Testing (White-box & Black-box testing)				May 1 - May 7
Bug Fixing, Performance Optimization & Security Auditing				May 8 - May 14
User Acceptance Testing (UAT) in specific Kigali Sectors				May 15 - May 21
Final Documentation, Thesis Compilation & Report Writing				May 22 - May 25

3.3.1 Phase Narratives

- **Phase 1 (Research):** Dedicated to strictly empirical data gathering. The team will interact directly with the Kigali public and authorities to ensure the software solves real problems, rather than assumed ones.
- **Phase 2 (Design):** Focuses heavily on the blueprint. Using tools like Figma, the team will ensure the Kinyarwanda/English interface is intuitive before a single line of code is written.
- **Phase 3 (Implementation):** The core coding phase. Utilizing rigorous collaborative programming techniques, the frontend and backend will be constructed simultaneously and joined via APIs.
- **Phase 4 (Delivery):** Prioritizes stability. The system will be deployed to a live cloud environment and tested by actual residents in sectors like Remera to gather real-world usage metrics before the final academic defense.

3.4. Estimated Budget

While the researchers will execute the software development as part of their academic requirements

(reducing labor costs), a modest budget is required for data collection, cloud hosting, and field operations.

Item / Resource	Description	Estimated Cost (RWF)
Internet & Cloud Hosting	Free AWS Credits	0
Field Data Collection	Transport to sectors, printing of consent forms, and minor survey incentives.	10,000
Software Assets	Any required premium UI icons or mapping API (Google Maps/Mapbox) credits.	20,000
Documentation & Binding	Printing and hard-binding of the final thesis books for the university.	5,000
Contingency Fund	10% set aside for unforeseen technical or logistical expenses.	10,000
Total Estimated Budget		45,000 RWF

4. Conclusion

Maintaining a sustainable, healthy, and sanitary urban environment relies fundamentally on highly coordinated solid waste management. As demonstrated by the synthesis of current global literature and the specific, pressing challenges faced locally at the Nduba dumpsite, traditional collection models relying on static schedules and manual oversight are no longer viable for rapidly expanding, dynamic cities like Kigali. Effective, modern SWM requires a fully integrated digital ecosystem featuring real-time monitoring, predictive strategic planning, and, most importantly, active and enthusiastic community participation.

This proposed web application directly addresses these glaring operational and research gaps by blending user-centric, mobile-first design with modern digital concepts such as crowdsourced GIS reporting and predictive routing logistics. By dramatically enhancing transparency between contractors and the public, eliminating operational redundancies for local collection fleets, and

empowering everyday citizens to act as active, vocal stakeholders in their city's hygiene, this digital intervention will provide the Rwandan community with a highly efficient, scalable, and sustainable framework for urban waste management, aligning perfectly with Kigali's vision of becoming Africa's premier Smart City.

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